

# Angela Qian

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## EDUCATION

**Purdue University** | *B.S. in Computer Science, Minor in Mathematics*

**Aug 2023 – May 2027**

## EXPERIENCE

**Undergraduate Research Assistant** | *CAMP Lab at Purdue University*

**Feb 2025 - Present**

- Co-authored a paper submitted to NeurIPS on Bayesian social deduction with graph-informed language models
- Conducted experiments involving human participants
- Assisted in developing and evaluating methods for hallucination detection

**Software Engineering Intern** | *Cepton Technologies Inc.*

**May 2025 - Aug 2025**

- Built in Rust a real-time object tracking pipeline using LiDAR-generated point cloud data on a per-frame basis
- Developed a 2D rectangle selection tool for a 3D point cloud viewer, enabling users to select points from any camera perspective and display their coordinates dynamically by transforming 3D point data using projection matrices
- Engineered a neural network based on U-Net segmentation to do real-time foreground-background segmentation masking, given depth and reflectivity maps

**Teaching Assistant** | *Purdue University Computer Science Department*

**Jan 2025 - Present**

- Led weekly PSO sessions, teaching problem-solving techniques and reinforcing theory taught in class
- Graded assignments, providing detailed feedback to help students improve and better understand material
- Held weekly office hours to clarify course material and guide students through homework challenges
- Conducted academic dishonesty investigations and interviewed students in coordination with course staff

**Undergraduate Researcher** | *Purdue University Vertically Integrated Projects*

**Aug 2024 – Dec 2024**

- Designed a program that uses convolution, the Laplacian filter, and the Sobel filter to detect image blur
- Developed algorithms involving Fourier Transforms to generate clear, focused photos from blurred images
- Presented in Purdue's 2024 Fall Undergraduate Research Expo

**Competition Mathematics Instructor** | *AlphaStar Academy*

**July 2024 – Aug 2024**

- Delivered lectures on Algebra, Number Theory, Geometry, and Counting/Probability
- Held review sessions with practice problems to prepare students for the AMC10, AMC12, and AIME exams
- Presented feedback to students and parents, highlighting focus areas for continued academic growth

## PUBLICATIONS

**Bayesian Social Deduction with Graph-Informed Language Models**

Shahab Rahimirad, Guven Gergerli, Lucia Romero, **Angela Qian**, Matthew Lyle Olson, Simon Stepputtis, Joseph Campbell (2025) *In submission for NeurIPS 2025*

## PROJECTS

**2048 AI Player** | *PyTorch, TensorFlow, Jupyter Notebook, JavaScript*

**June 2025 - Present**

- Developed a neural network to play 2048 using imitation learning trained on a custom dataset of my own gameplay
- Built model in PyTorch to predict actions from board states, and exported it via ONNX to deploy in TensorFlow.js
- Currently extending the project with reinforcement learning to fine-tune decision making through self-play

**ScribbleScore Handwriting Improvement** | *Python, OpenCV, Numpy, React, Flask*

**Feb 2025**

- Worked in a group of 4 over the course of a 36-hour hackathon (Boilermake XII)
- Designed bounding-box logic to group words and measure spacing consistency across clusters
- Implemented Sobel edge detection, line of best fit analysis to assess writing straightness, extracting text from images, and computing score

## SKILLS

**Languages:** English (Native), Mandarin Chinese (Native)

**Programming:** Python, Rust, C/C++, JavaScript, Typescript, x86\_64 Assembly, SQL, MongoDB, HTML, CSS, Java

**Developer Tools:** Git, GitHub, VS Code, Eclipse, Jupyter Notebook, Linux, Overleaf, React, LaTeX, VIM, MySQL, Neo4J, Next.js, TailwindCSS, Agile & Scrum

**Libraries:** OpenCV, Matplotlib, PyTorch, TorchVision, TensorFlow, Pandas, NumPy